HW#2

MAE 140

Winter 2014

2–25 Find v_x and i_x in Figure P2–25.

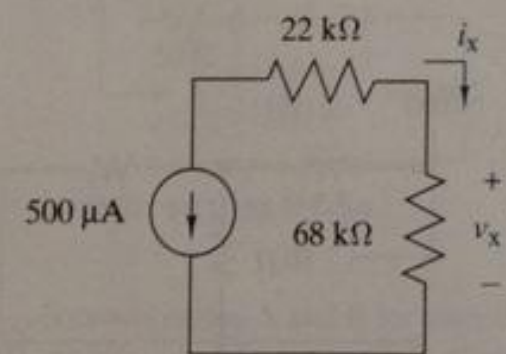

FIGURE P2–25

2–28 A modeler wants to light his model building using miniature grain-of-wheat light bulbs connected in parallel as shown in Figure P2–28. He uses two 1.5-V "C-cells" to power his lights. He wants to use as many lights as possible, but wants to limit his current drain to 500 µA to preserve the batteries. If each light has a resistance of 36 kΩ, how many lights can he install and still be under his current limit?

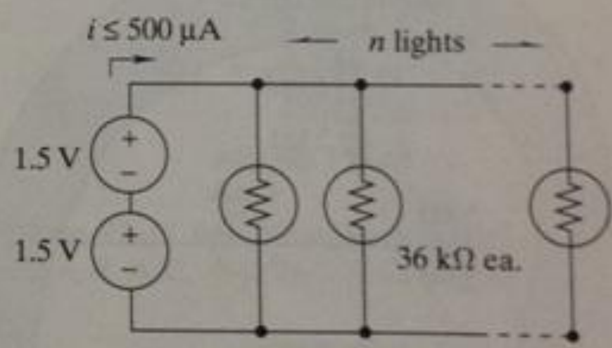

FIGURE P2–28

2–32 Figure P2–32 shows a subcircuit connected to the rest of the circuit at four points.

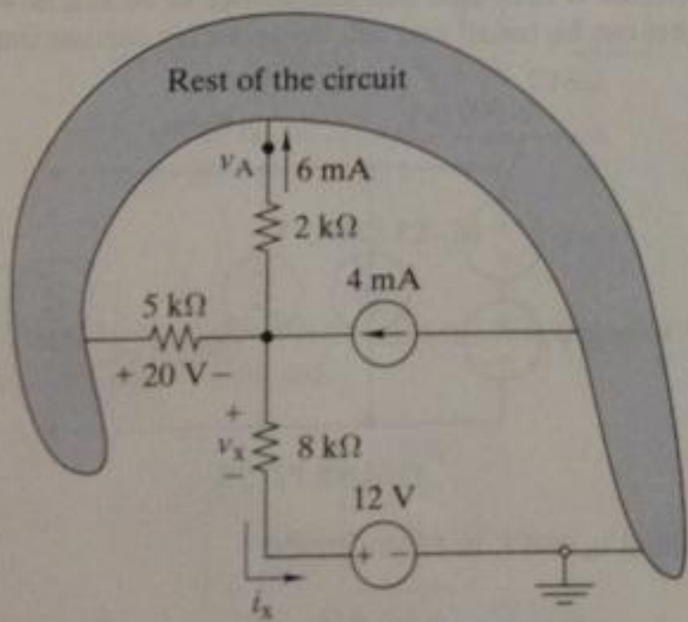

FIGURE P2–32

(a) Use element and connection constraints to find v_x and i_x.
(b) Show that the sum of the currents into the rest of the circuit is zero.
(c) Find the voltage v_A with respect to the ground in the circuit.

2–36 Find the equivalent resistance R_{EQ} in Figure P2–36.

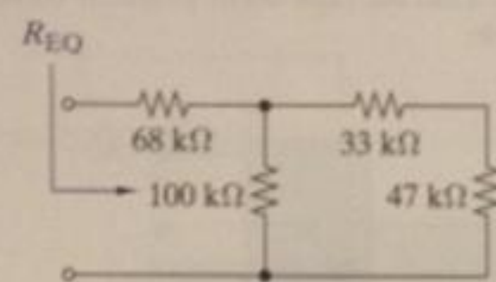

FIGURE P2–36

2–38 Equivalent resistance is defined at a particular pair of terminals. In the following figure the same circuit is looked at from two different terminal pairs. Find the equivalent resistances R_{EQ1} and R_{EQ2} in Figure P2–38. Note that in calculating R_{EQ2} the 33-kΩ resistor is connected to an open circuit and therefore doesn't affect the calculation.

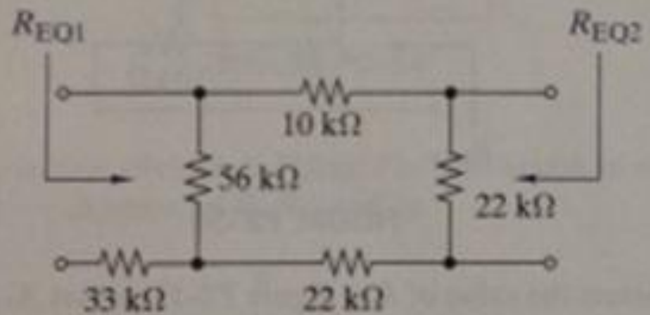

FIGURE P2–38

2–43 In Figure P2–43 find the equivalent resistance between terminals A-B, A-C, A-D, B-C, B-D, and C-D.

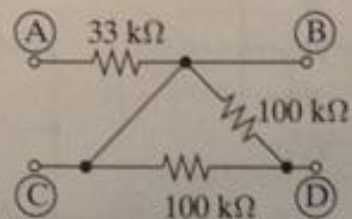

FIGURE P2–43

2–55 Use voltage division in Figure P2–55 to obtain an expression for v_L in terms of R, R_L, and v_S.

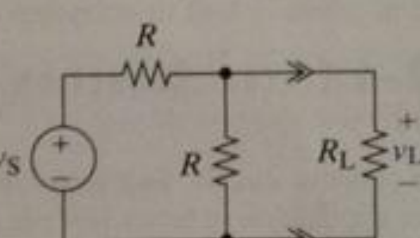

FIGURE P2–55

2–60 **A** The 1-kΩ load in Figure P2–60 needs 5 V across it to operate correctly. Where should the wiper on the potentiometer be set (R_x) to obtain the desired output voltage?

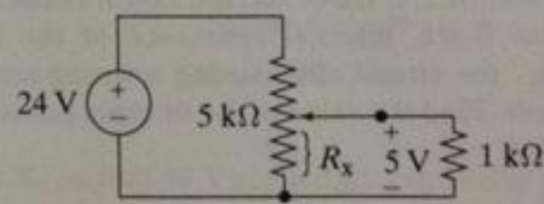

FIGURE P2–60

2–70 Use circuit reduction to find v_x and i_x in Figure P2–70.

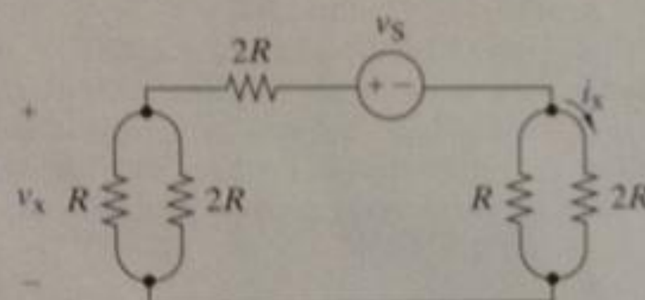

FIGURE P2–70

2–73 Use source transformation to find i_x in Figure P2–73.

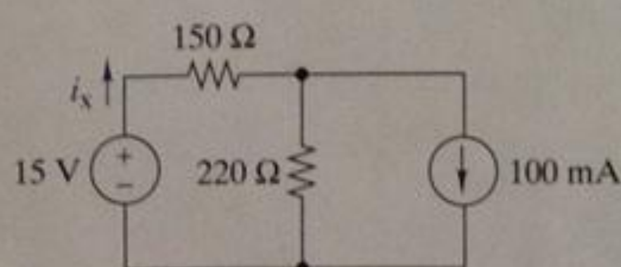

FIGURE P2–73